Roll No. | | | | | | | |

End Semester Examination 2025

Name of the Course: M.Tech. Biomedical Engineering **Semester: I**
Name of the Paper: Biomaterials Engineering **Paper Code: BEM 101**

Time: 3 Hours **Maximum Marks: 100**

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	**CO3**
(a)	Explain the chemical structure, mechanical characteristics, and biocompatibility of polyamides utilized in medical implant applications.	
(b)	Discuss the significance and applications of acrylic polymers in orthopedic and dental implants.	
(c)	Define fluorocarbon polymers. Describe their key properties and explain how these properties enhance the long-term performance of biomedical implants.	
Q2	(20marks)	**CO1/CO2**
(a)	Explain the various types of biomaterials — metals, polymers, ceramics, and composites — and compare their key properties relevant to biomedical applications.	
(b)	Describe the surface, physical, and mechanical characteristics of biomaterials, and discuss their significance in the design of biomedical implants.	
(c)	Outline the materials used, essential design considerations, and major applications of orthopedic and dental implants.	
Q3	(20marks)	**CO3/CO5**
(a)	Define bioceramics and explain the various types of bioceramic materials used in biomedical applications, providing suitable examples.	
(b)	Describe polymer-based composite materials containing osteogenic fillers and discuss their applications in bone tissue engineering.	
(c)	Compare the key properties and biomedical applications of bioceramic materials and composite biomaterials.	
Q4		**CO3**
(a)	Describe the key characteristics of acetal polymers (polyoxymethylene) and discuss the reasons for their suitability in joint replacement applications.	
(b)	Explain the properties, benefits, and biomedical uses of polyolefins as materials for medical implants.	
(c)	Classify polymeric implant materials into thermoplastics, thermosets, and elastomers, and provide suitable examples for each category.	
Q5		**CO5**
(a)	Define biocompatibility and distinguish between blood compatibility and tissue compatibility, providing suitable examples.	
(b)	Discuss the key factors that influence the biocompatibility of implant materials.	
(c)	Differentiate between *in vitro* and *in vivo* testing of biomaterials and explain the importance of each testing method.	